

The empirical recurrence for OEIS A236889: recovering a conjecture that is not written down

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7 September 2026

Abstract

OEIS A236889 records “Empirical recurrence of order 59”, with the recurrence itself in a linked file rather than in the entry. The conjecture can nevertheless be settled, and the recurrence recovered, without reading that file. The entry counts arrays under a condition local to a bounded window of lines, so the count is a walk count on a finite digraph and satisfies some monic recurrence of order at most the number of its states with distinct futures, here $S' = 525$; Berlekamp–Massey applied to $2S'$ exact terms therefore returns the MINIMAL such recurrence. Its order is exactly 59, the order the entry states, and the same is true of every tail of the sequence. Any recurrence of order 59 that the sequence satisfies from any point on must then have a characteristic polynomial that is a multiple of the minimal one and of the same degree, hence equal to it. So the entry’s recurrence is the one displayed here, whatever its file contains, and it is proved.

2020 Mathematics Subject Classification. 05A15, 11B37, 15A18.

1 A conjecture one cannot read

OEIS A236889 is “Number of $(n+1) \times (4+1)$ 0..2 arrays with the maximum plus the upper median minus the lower median minus the minimum of every 2×2 subblock equal.”. It has offset 1 and begins

2605, 19005, 134361, 1107653, 9168057, 83876241, 790250057, 8009517233, ...

The entry states:

Empirical recurrence of order 59 (see link above)

As of the “Last modified” line on the live entry (Jul 23 2025 09:38:40, revision 7) this is still recorded as empirical. The recurrence’s coefficients are not in the entry text.

2 The count is a walk count

The entry’s condition is local: it constrains a bounded window of consecutive lines of the array and nothing further. The admissible configurations of one such window are the vertices of a finite digraph, an edge joining u to v when v may follow u , and an admissible array is exactly a walk. Hence, with M the adjacency matrix on $S = 525$ vertices, $a(n) = \iota^\top M^{j(n)} \tau$ for a linear reindexing j fixed by the entry’s shape and checked against its published terms. By the Cayley–Hamilton theorem M^S is an integer combination of $I, M, \dots, M^{S-1}$, so:

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S' = 525$.*

3 Recovering the recurrence

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S is determined, together with its minimal recurrence, by its first $2S$ terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Running it on $2S = 1050$ exact terms of the model returns order 59 — the order the entry states — with the integer coefficients

$$a(n) = \sum_{i=1}^{59} c_i a(n-i),$$

c_1	34	c_{16}	−48792567747598	c_{31}	−365471222225375015802	c_{46}	−1552780744624199
c_2	−185	c_{17}	−1273250810321394	c_{32}	−704020130154002853708	c_{47}	−708417479956346
c_3	−5908	c_{18}	2091378305419122	c_{33}	1227559759371963405313	c_{48}	6779894729826194
c_4	74552	c_{19}	14297043960301944	c_{34}	1153301386092751323901	c_{49}	1528359118957540
c_5	283328	c_{20}	−37286467578392238	c_{35}	−2919638712388688583199	c_{50}	−202102717863494
c_6	−8374817	c_{21}	−112574339537813336	c_{36}	−1136597959439943060255	c_{51}	−162704919289130
c_7	10534088	c_{22}	428783446748993194	c_{37}	5070474820265279613031	c_{52}	3993838772698607
c_8	489521682	c_{23}	577687154536001930	c_{38}	183737744140242040957	c_{53}	−55943010760294
c_9	−1906530473	c_{24}	−3490262240749185181	c_{39}	−6495032196767970530335	c_{54}	−49096886915899
c_{10}	−16972963511	c_{25}	−1346137842150490438	c_{40}	1387747306228130080418	c_{55}	370782569965289
c_{11}	107718421125	c_{26}	20787518220674525927	c_{41}	6132105128845136837590	c_{56}	329838687210700
c_{12}	345372927901	c_{27}	−5485667019030258476	c_{42}	−2498784266174149912918	c_{57}	−3747788749799
c_{13}	−3600389856595	c_{28}	−91609792610456615156	c_{43}	−4226951401351241976662	c_{58}	−8756083556352
c_{14}	−2900730264051	c_{29}	70322065479111273108	c_{44}	2432009107275432501552	c_{59}	11839211569152
c_{15}	80759918251879	c_{30}	298070019535036474341	c_{45}	2085062351252950727160		

Theorem 3. *The recurrence above holds for every n , and it is the recurrence the entry records.*

Proof. It holds by Lemma 2, which returns a recurrence the sequence satisfies from its first term. For the identification: the entry asserts that the sequence satisfies SOME recurrence of order 59, possibly only from some index on. Let q be that recurrence’s characteristic polynomial and $q_{\min}$ the minimal polynomial of the tail of the sequence. Then $q_{\min} \mid q$. Berlekamp–Massey run on the sequence with its first $59 + 4$ terms removed returns order 59 as well, so $\deg q_{\min} = 59 = \deg q$; both are monic, so $q = q_{\min}$, which is the polynomial computed above. $\square$

4 Verification

Three checks.

First, the walk model reproduces all 15 terms the entry publishes, exactly, in integer arithmetic. This is what ties the model to the entry, and it is the only place where reading the entry’s English enters.

Second, the recovered recurrence was evaluated directly on those published terms, with no matrices and no Berlekamp–Massey involved, and holds at every index where the entry gives enough earlier terms to test it.

Third, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once in exact rational arithmetic; and once more on the sequence with its first $59 + 4$ terms removed, which is what the identification above requires. All three agree.

Remark 4. What is unusual here is that the conjecture’s statement was never read. The entry pins it down to a one-parameter family — the recurrences of order 59 that this sequence satisfies — and that family turns out to have exactly one member.

References

- [1] The OEIS Foundation, *The On-Line Encyclopedia of Integer Sequences*, <https://oeis.org>, 2026. Sequence A236889.
- [2] J. L. Massey, Shift-register synthesis and BCH decoding, *IEEE Trans. Inform. Theory* **15** (1969), 122–127.
- [3] R. P. Stanley, *Enumerative Combinatorics*, Volume 1, 2nd ed., Cambridge University Press, 2011. (Transfer-matrix method, Section 4.7.)
- [4] R. A. Horn and C. R. Johnson, *Matrix Analysis*, 2nd ed., Cambridge University Press, 2013.